

Unit 08: Number, Ranking and Time Sequence

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Objective

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- We will understand different types of questions on Number Test.
- We will understand different types of questions on Ranking Test.
- We will understand different types of questions on Time Sequence Test.

Introduction

Number-Ranking-Time Sequence section has all the different types of questions on Number Test, Ranking Test, and Time Sequence Test. In this type of questions, generally, you are given a long series of numbers. The candidate is required to find out how many times a number satisfying the conditions, specified in the question, occurs.

1 Minute = 60 seconds

1 Hour = 60 minutes

1 Day = 24 hours

1 Week = 7 days

1 Month = 4 weeks

1 Year = 12 months

1 Ordinary year = 365 days

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1 Leap year = 3 66 days

1 Century = 100 years.

Month Name	Total number of days
January	31
February	28
March	31
April	30
May	31
June	30
July	31
August	31
September	30
October	31
November	30
December	31.

Other facts to be remembered

A day is the period of the earth's revolution on its axis.

A 'Solar year' is the time taken the earth to travel round the sun. It is equal to 365 days, 5 hours, 48 minutes and 47- seconds nearly.

A 'Lunar month' is the time taken the moon to travel round the earth. It is equal to nearly 28 days.

Leap Year

If the number of a given year is divisible by 4, it is a leap year. Hence, the years like 1996, 2008, 2012 are leap years. But years like 1997, 1991, 2005, 2007 are not divisible by 4 and therefore, such years are not leap years.

In a leap year, February has 29 days.

A leap year has 52 weeks and 2 days.

8.1 Number Test

In this type of question, generally, a set group or series of numerals is given and the candidate is asked to trace out numerals following certain given conditions or lying at specifically mentioned positions after shuffling according to a certain given pattern. In these types of questions, a number, a set of numbers, series of digit is given and the candidate is asked to trace out digit following certain given conditions. In these sort of problems, a number, an arrangement of numbers, arrangement of digit is given and the applicant is requested that follow out digit taking after certain given conditions or lying at particular said positions in the wake of rearranging as indicated by a specific given example.



Example 1. How many such 5s are there in the following number sequence each of which is immediately preceded by 3 or 4 but not immediately followed by 8 or 9?

3 5 9 5 4 5 5 3 5 8 4 5 6 7 3 5 7 5 5 4 5 2 3 5 1 0

(A) None

(B) Three

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- (C) Four
(D) Five
(E) None of these

Solution:

As you know, a number that comes after a given number is said to follow it while the one which comes before the given number precedes it.

Thus, the number satisfying the given conditions may be marked as follows:

3 5 9 5 4 5 3 5 8 4 6 7 3 7 5 5 4 2 3 1 0

Clearly, there are five such 5s. Hence, the answer is (d).



Example2. What will be last digit of the 3rd number from top when the numbers given below are arranged in descending order after reversing the position of the digits within each number?

517 325 639 841 792

- (a) 2
(b) 5
(c) 7
(d) 3

Sol. (d) The given numbers are:

517 325 639 841 792

After reversing, the numbers becomes as follows: 715 523 936 148 297

When arranged in descending order the numbers become as follows:

936 715 523 297 148;

Now, the third number from top is 523. Hence, the last digit of 523 is 3.

∴ Option (d) is correct.



Example 3 How many 5s are there in the following number sequence which are immediately preceded by 7 and immediately followed by 6?

Terms : 7 5 5 9 4 5 7 6 4 5 9 8 7 5 6 7 6 4 3 2 5 6 7 8

- (a) 1
(b) 2
(c) 3
(d) 4

Solution. (a) Here, 7 5 5 9 4 5 7 6 4 5 9 8 7 5 6 7 6 4 3 2 5 6 7 8

Preceded by 7 and followed by 6 So, there is only one such 5.

738 429 156 273 894



Example: Which of the following will be the second digit of the third number from the top when they are arranged in descending order, after the first digit in each number is changed to its next higher digit?

- A) 2

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B) 3

C) 5

D) 7

E) 9

Answer:

Option A

Explanation:

The new numbers formed are : 838, 529, 256, 373, 994.

These, in descending order, are : 994, 838, 529, 373, 256.

The third number from the top is 529, and its second digit is 2.

The digits of each of the following five numbers are written in reverse order and five new number are obtained

513 726 492 865 149



Example: Which of the following will be the middle digit of the third number from the top when the new numbers are arranged in descending order?

A) 1

B) 2

C) 6

D) 9

E) None of these

Answer:

Option C

Explanation:

The new numbers, arranged in descending order are 941 627 568 315 294 .

The third number from the top is 568 and its middle digit is 6.



Example: In the following sequence of instructions, 1 stands for Run, 2 stands for Stop, 3 stands for Go, 4 stands for Sit and 5 stands for Wait. If the sequence were continued, which instruction will come next

4 4 5 4 5 3 4 5 3 1 4 5 3 1 2 4 5 4 5 3 4 5 3

A) Wait

B) Sit

C) Go

D) Stop

E) Run

Answer:

Option E

Explanation:

The given sequence may be analysed as under:

4 / 45 / 453 / 4531 / 45312 / 45 / 453 / 453

Following the above sequence, the next number is 1 which stands for 'Run'

8.2 Ranking Test

In this type of questions, generally the ranks of a person both from the top and from the bottom are mentioned and the total number of persons is asked. However, sometimes this question is put in the form of a puzzle of interchanging seats by two persons. In these sort of questions, generally the rank of a person both from the top and from the base are said and the aggregate number of persons is asked. In any case, in some cases this grouping is placed as a riddle of interchanging seats by two persons. In these types of questions, generally the ranks of a person both from the top and from the bottom are mentioned and the total number of persons is asked. When we are doing number and ranking we know basically two type of symbols first is greater than(>)and the other one is less than(<).To solve these questions we only know the three things:

Symbol of Greater Than and Less than means if $A > B$ IMPLIES THAT (A is greater than B)or $B < A$ (B is smaller than A). These are for the same person:

$$\text{TOTAL} = \{(\text{TOP} + \text{BOTTOM}) - 1\}$$

$$\text{TOTAL} = \{(\text{LEFT} + \text{RIGHT}) - 1\}$$

In numbering and ranking arrangement questions, position/rank of a person from left-right/top-bottom of a row/class is to be determined or rank/position is given & total no. of persons is to be calculated. You may also be asked to determine, using data given, which floor which person lives on.



Note:

- 1) Read the statement line by line and apply the cases as explained below.
- 2) Position can be from either sides of row and rank is always from top or bottom of the row

One of the type of Number Ranking.

- 1) Total number of persons = $\{(\text{sum of positions of same person from both sides i.e. left and right side}) - 1\}$

OR

- 2) Position of a person from opposite side = $\{(\text{Total no. of persons} - \text{Position of same person from given side}) + 1\}$



Example:- In a row of persons, position of A from left side of the row is 27th and position of A from right side of the row is 34th. Find total no. of persons in the row?

Solution:

Total no. of students = (Position of A from left + Position of A from right) - 1

$\Rightarrow$ Total no. of students = $(27 + 34) - 1 = 61 - 1 = 60$.



Example 4 : A class of boys stands in a single line. One boy is nineteenth in order from both the ends. How many boys are there in the class?

- (a) 27
- (b) 37
- (c) 38

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(d) 39

Solution. (b) Clearly, number of boys in the row = $(18 + 18 + 1) = 37$ The question can be solved by formula also, total number of boys in the row = $19 + 19 - 1 = 37$

Formula Total number of persons in a row or class = (Rank of a person from upper end or left end) + (Rank of that person from lower or right end) - 1



Example:- Rohan ranks seventh from the top and twenty-sixth from the bottom in a class. How many students are there in the class?

(A) 31

(B) 32

(C) 33

(D) 34

Solution:

Clearly, the whole class consists of:

(i) 6 student who have ranks higher than Rohan;

(ii) Rohan; and

(iii) 25 students who have ranks lower than Rohan, i.e, $(6+1+25) = 32$ students. Hence, the answer is (b).

In such problems, the ranks of a person both from the top from the bottom are given and on the basis of this the total number of persons is asked. Sometimes question is twisted also and position of a particular person is asked.



Example 5. Karishma ranks 10th from the top and 15th from the bottom in an examination. Find the total number of students in Karishma's class.

(a) 35

(b) 31

(c) 28

(d) 30

Sol. (d) As per the question; the class has

(i) 15 students higher than Karishma

(ii) 14 students lower than Karishma

(iii) Karishma

 $\therefore$ Total number of students = $15 + 14 + 1 = 30$ Hence, option (d) is correct.

Example 6f:- In a row of trees, one tree is fifth from either end of the row. How many trees are there in the row ?

Option:

A. 8

B. 9

C. 10

D. 11

Answer: B . 9

Justification: Clearly, number of trees in the row = $(4 + 1 + 4) = 9$.

8.3 Time Sequence Test



Example.1. Satish remembers that his brother's birthday is after fifteenth but before eighteenth of February whereas his sister Kajal remembers that her brother's birthday is after sixteenth but below nineteenth of February. On which day in February is Satish's brother's birthday?

- (A) 16th
- (B) 17th
- (C) 18th
- (D) 19th
- (E) None of these

Solution:

According to Satish, the brother's birthday is on one of the days among 16th and 17th February.

According to Kajal, the brother's birthday is on one of the days among 17th and 18th February.

Clearly, Satish's brother's birthday is on the day common to both the above groups, i.e., 17th February.



Example.7. Neena returned home after 3 days earlier than the time she had told her mother. Neena's sister Veena reached five days later than the day Neena was supposed to return. If Neena returned on Thursday, on what day did Veena return?

- (a) Friday
- (b) Saturday
- (c) Wednesday
- (d) Sunday

Sol. (a) Neena returned home on Thursday. Neena was supposed to return 3 days later, i.e., on Sunday.

Veena returned five days later from Sunday, i.e., on Friday.

$\therefore$ Option (a) is the correct option.



Example 8:- If the seventh day of a month is three days sooner than Friday, what day will it be on the nineteenth day of the month.

Solution:

Given that seventh day of a month is three days sooner than Friday. So, the seventh day of the month is three days earlier than Friday, which is Tuesday. Therefore, the fourteenth day is also Tuesday. Hence, nineteenth day is Sunday.



Example 9. Vandana remembers that her father's birthday is between 13th and 16th of June. Whereas her brother remembers that their Father's birthday is between 14th and 18th of June. On which day is their Father's birthday?

- (a) 14th June
- (b) 16th June

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(c) 15th June

(d) 18th June

Sol. (c) According to Vandana her father's birthday is on one of the days among 14th and 15th June. Vandana's brother, the father's birthday is one of the days among 15th 16th and 17th June.

It is obvious that the father's birthday is on the day to both the above groups. The common day is 15th. Hence, the father's birthday falls on 15th June. 1 Option (c) is the correct option.



Example 10:- How many such 6's are there in the following number sequence each of which is immediately preceded by 3 or 4 but not immediately followed by 8 or 9?

3 6 9 6 4 6 6 3 6 8 4 6 5 7 3 6 7 6 6 4 6 2 3 6 1 0

Solution:

Given number sequence is

3 6 9 6 4 6 6 3 6 8 4 6 5 7 3 6 7 6 6 4 6 2 3 6 1 0

A number which comes after a given number is said to follow it while the one which precedes the given number goes before it.

Thus, the numbers satisfying the given conditions may be marked (with red colour) as follows:

3 6 9 6 4 6 6 3 6 8 4 6 5 7 3 6 7 6 6 4 6 2 3 6 1 0

Therefore, there are five such 6's.



Example 11:- How many pairs of successive numbers have a difference of 3 each? If the series:

7 4 1 2 2 8 4 1 2 1 8 5 5 2 2 1 7 1 4 2 1 4 2 6 3.

Solution:

Given series is:

7 4 1 2 2 8 4 1 2 1 8 5 5 2 2 1 7 1 2 1 4 2 6 3.

Here,

$$7 - 4 = 3$$

$$4 - 1 = 3$$

$$8 - 5 = 3$$

$$5 - 2 = 3$$

$$4 - 1 = 3$$

$$6 - 3 = 3$$

So, the pairs of successive numbers having a difference of 3 can be shown below:

7 4 1 2 2 8 4 1 2 1 8 5 5 2 2 1 7 1 2 1 4 2 6 3

Therefore, there are six such pairs.

Summary

The key concepts learned from this unit are: -

- We have learnt how to solve different types of questions on Number Test.
- We have learnt how to solve different types of questions on Ranking Test.
- We have learnt how to solve different types of questions on Time Sequence Test.

Keywords

- Series of numbers.
- Number sequence
- Number Test.
- Ranking Test.
- Time Sequence Test.

Self Assessment

1. How many 3's are there in the following sequence which is neither preceded by nor immediately followed by 9?

9 3 6 6 3 9 5 9 3 7 8 9 1 6 3 9 6 3 9

- A. One
B. Two
C. Three
D. None of these

2. Count each 7 which is not immediately preceded by 5 but is immediately followed by either 2 or 3. How many such 7's are there?

5 7 2 6 5 7 3 8 3 7 3 2 5 7 2 7 3 4 8 2 6 7 8

- A. 2
B. 3
C. 4
D. 5

3. How many 7's are there in the following series which are preceded by 6 which is not preceded by 8?

8 7 6 7 8 6 7 5 6 7 9 7 6 1 6 7 7 6 8 8 6 9 7 6 8 7

- A. Nil
B. One
C. Two
D. None of these

4. In a queue, Amrita is 10th from the front while Mukul is 25th from behind and Mamta is just in the middle of the two. If there were 50 persons in the queue, what position does Mamta occupy from the front?

- A. 20th
B. 19th
C. 18th
D. 17th

5. Manisha ranked sixteenth from the top and twenty-ninth from the bottom among those who passed an examination. Six boys did not participate in the competition and five failed in it. How many boys were there in the class?

- A. 40
B. 44
C. 50
D. 55

6. Some boys are sitting in a row. P is sitting fourteenth from the left and Q is seventh from the right. If there are four boys between P and Q, how many boys are there in the row?

- A. 25
B. 23
C. 21
D. 19

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7. A bus for Delhi leaves every thirty minutes from a bus stand. An enquiry clerk told a passenger that the bus had already left ten minutes ago and the next bus will leave at 9.35 a.m. At what time did the enquiry clerk give this information to the passenger?
 A. 9.10 a.m.
 B. 8.55 a.m.
 C. 9.08 p.m.
 D. 9.15 a.m.
8. Sangeeta remembers that her father's birthday was certainly after eighth but before thirteenth of December. Her sister Natasha remembers that their father's birthday was definitely after ninth but before fourteenth of December. On which date of December was their father's birthday? (Bank P.O. 1998)
 A. 10th
 B. 11th
 C. 12th
 D. Data inadequate
9. The train for Lucknow leaves every two and a half hours from New Delhi Railway Station. An announcement was made at the station that the train for Lucknow had left 40 minutes ago and the next train will leave at 18.00 hrs. At what time was the announcement made?
 A. 15.30 hrs
 B. 15.50 hrs
 C. 16.10 hrs
 D. 17.10 hrs
10. In the series given below, count the number of 9's, each of which is not immediately preceded by 5 but is immediately followed by either 2 or 3. How many such 9's are there?
 1 9 3 2 1 7 4 2 6 9 7 4 6 1 3 2 8 7 4 1 3 8 3 2 5 6 7 4 3 9 5 8 2 0 1 8 7 4 6 3
 A. one
 B. three
 C. five
 D. six

Answers for Self Assessment

- | | | | | |
|------|------|------|------|-------|
| 1. D | 2. A | 3. D | 4. C | 5. D |
| 6. A | 7. D | 8. D | 9. C | 10. B |

Review Questions

1. Pointing to Ajay, Radha said, "His father is the only son of my grandfather". How is Radha Related to Ajay?
2. How many 4's are there preceded by 7 but not followed by 3? 5 9 3 2 1 7 4 2 6 9 7 4 6 1 3 2 8 7 4 1 3 8 3 2 5 6 7 4 3 9 5 8 2 0 1 8 7 4 6 3.
3. Rohan ranks 7th from the top and 26th from the bottom in the class. How many students are there in the class?
4. Manik is 14 th from the right end in the row of 40 students . What is his position from the left end ?
5. In a row of boys facing the north , A is 16th from the left end and C is 16 th from the right end . B , who is 4th to the right of A , is 5th to the left of C, in a row. How many boys are there in a row?
6. In the following series of numbers, find out how many times, 1, 3 and 7 have appeared together, 7 being in the middle and 1 and 3 on either side of 7 ?
 2 9 7 3 1 7 3 7 7 1 3 3 1 7 3 8 5 7 1 3 7 7 1 7 3 9 0 6

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7. If the numbers from 5 to 85 which are exactly divisible by 5 are arranged in descending order, which would come at the eleventh place from the bottom ?
8. In a row of students, Anil is 7th from left, while Sunil is 18th from right. Both of them interchanged their positions such that Anil becomes 21st from left. What will be the total number of students in the class?
9. Mohan is taller than Shyam but shorter than Ramesh. Ramesh is taller than Rajat but shorter than Gautam. If Shyam is taller than Rajat, then who is the shortest among all?
10. Sunita leaves her house at 20 min to seven in the morning, reaches Vineeta's house in 25 min, they finish their breakfast in another 15 min and leave for their office which takes another 35 min. At what time, did they leave Vineeta's house to reach their office?

**Further Readings**

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing
3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle
4. Analytical Reasoning By M.K. Pandey, Banking Service Chronicle

